(a) Finding a_k for Fourier Series

Waveform (1): ~~Square~~

(1) a_0

$$a_0 = \frac{1}{T}\int_{-T/2}^{T/2} f(t)\,dt = \frac{1}{2}\int_{-0,5}^{0,5} 1\,dt = 0,5$$

(2) a_k

$$a_k = \frac{2}{T}\int_{-0,5}^{0,5} \cos(k\pi t)\,dt = \frac{\sin(k\pi/2)}{k\pi}$$

$$\boxed{a_k = \begin{cases} 0,5 & k=0 \\ \dfrac{\sin(k\pi/2)}{k\pi} & k\neq 0 \end{cases}}$$

Waveform (2):

(1) a_0

$$a_0 = \frac{1}{T}\left(\int_{-0,5}^{0,5} 1\,dt + \int_{3,5}^{4,5} 1\,dt\right) = \frac{2}{4} = 0,5$$

(2) a_k

$$a_k = \frac{2}{4}\left(\int_{-0,5}^{0,5} \cos\left(\frac{k\pi}{2}t\right)dt + [\text{Shifted integral}]\right) =$$

$$= \frac{2\sin(k\pi/4)}{k\pi}$$

$$\boxed{a_k = \begin{cases} 0,5 & k=0 \\ \dfrac{2\sin(k\pi/4)}{k\pi} & k\neq 0 \end{cases}}$$

(b)

Waveform (1):

0) $a_0 = 0,5$

1) $a_{\pm1} = \dfrac{\sin(\pm\pi/2)}{\pm\pi} = \pm\dfrac{1}{\pi} \approx \pm 0,318$

2) $a_{\pm2} = 0$ as all the even values of k cause $\sin = 0$ => so all (k = even) = 0

3) $a_{\pm3} = \dfrac{\sin(\pm\frac{3\pi}{2})}{\pm3\pi} = \dfrac{\mp1}{3\pi}$

4) $a_{\pm4} = 0$

5) $a_{\pm5} = \dfrac{\sin(\frac{5\pi}{\pm2})}{\pm5\pi} = \dfrac{\pm1}{5\pi}$

6) $a_{\pm6} = 0$

7) $a_{\pm7} = \dfrac{\sin(\frac{\pm7\pi}{2})}{\pm7\pi} = \mp\dfrac{1}{7\pi}$

8) $a_{\pm 8} = 0$

9) $a_{\pm 9} = \dfrac{\sin\left(\frac{9\pi}{2}\right)}{\pm 9\pi} = \pm \dfrac{1}{9\pi}$

10) $a_{10} = 0$

Waveform(2)

$a_0 = 0{,}5$

$a_{\pm 1} = \pm \dfrac{2\sin(\pi/4)}{\pi} \approx \pm 0{,}45$

$a_{\pm 2} = \pm \dfrac{2\sin(\pi/2)}{2\pi} \approx \pm 0{,}32$

$a_{\pm 3} = \pm \dfrac{2\sin(3\pi/4)}{3\pi} \approx \pm 0{,}15$

$a_{\pm 4} = \pm \dfrac{2\sin(4\pi/4)}{4\pi} = 0$

$a_{\pm 5} = \pm \dfrac{2\sin(5\pi/4)}{5\pi} \approx \mp 0{,}09$

$a_{\pm 6} = \pm \dfrac{2\sin(3\pi/2)}{6\pi} \approx \mp 0{,}106$

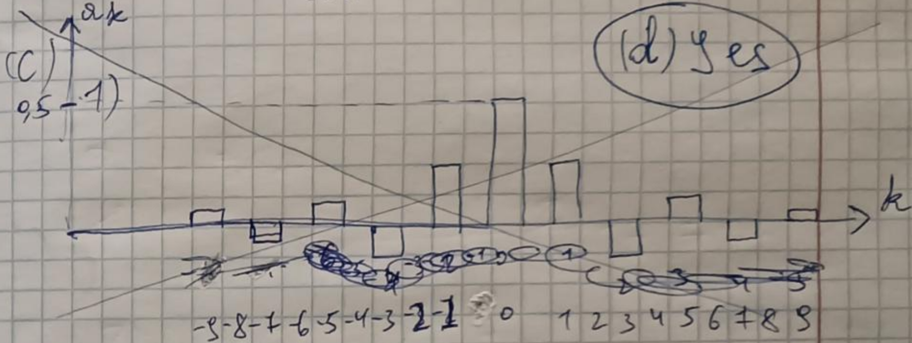
$a_{\pm 7} = \pm \dfrac{2\sin(7\pi/4)}{7\pi} \approx \pm 0{,}064$

$a_{\pm 8} = \pm \dfrac{2\sin(8\pi/4)}{8\pi} = 0$

$a_{\pm 9} = \pm \dfrac{2\sin(9\pi/4)}{9\pi} \approx \pm 0{,}050$

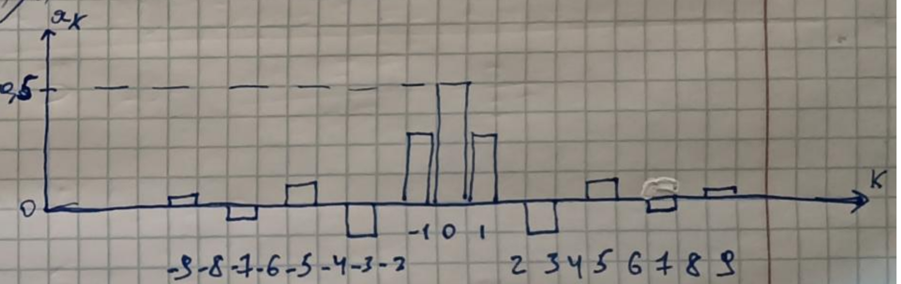
$a_{\pm 10} = \pm \dfrac{2\sin(10\pi/4)}{10\pi} \approx \pm 0{,}064$

(c) 1)



(d) yes

(c) 1)



2)